Honeywell Cable Ring Optimization

Product Requirements Document | Correctness-first algorithm research and evaluation

Version 1.0 • 23 October 2026 • Scope: optimization engine, experimentation and reproducibility only

1. Executive decision

Build a mathematically faithful optimizer for joint control-panel placement, eligible-room assignment, node sequencing and closed-loop cable routing. Select the best validated algorithm or algorithm portfolio by lexicographic feasibility and performance: first minimum hub count H, then minimum routing cost J. Do not predetermine that RL must win.

Decision: rebuild or repair the geometry/constraint foundation first; reuse verified modules and historical data as assets; rerun benchmarks whenever underlying semantics change. UI, dashboards and presentation polish are explicitly deferred.

2. Source of truth and official mathematical model

Authoritative source: supplied 8-slide “Cable Ring Challenge - Problem Statement4(1).pptx”, especially slides 2–8. Mentor Q&A is secondary; conflicts must be logged and resolved against the written rules/checker rather than assumed away.

Requirement	Acceptance interpretation
Eligible nodes	One fixed input node in each eligible living, bedroom, kitchen or storage room; no moving input nodes.
Hub capacity	At most 100 m^2 assigned eligible-room area AND at most 5 nodes.
Zone connectivity	Assigned eligible rooms form one connected territory via doors/open passages; walls, windows and entrance do not establish adjacency.
Hub placement	Within an assigned eligible room; ≥0.5 m from walls; may share a room with a node.
Ring	One closed nonbranching H→N1→…→Nn→H route per hub; each assigned node visited once.
Routing	Axis-aligned, inside inner boundary, no wall/window/external crossings; between rooms only through doors/open passages. Single-door retracing permitted.
Objective	Minimize lexicographically (H, J, area-gap tie-break), J = L + 5X + O + 0.5B; 1 mm tolerance.
Units	Per-plan scale = sqrt(inner.area / area); convert geometry before area, distance and clearance checks.
Output	Per-plan valid JSON with hubs, assignments, node sequence, full polylines, return leg and independently recomputed metrics.

Specification nuance: slides describe approximately 90% benchmark filtering and reference-solver feasibility, whereas the raw 17,000 ResPlan records include anomalous and potentially infeasible cases. Distinguish raw-data processing from official benchmark eligibility. Mentor suggestion about exceptional

excluded-area traversal is not an unconditional permission; treat as unresolved exception requiring formal evidence.

3. What exists today: assets and measured history

Asset / evidence	Reuse decision
17,000 ResPlan records; inventory, split manifests, anomaly audits	Reuse raw source and audit scripts; regenerate derived splits/eligibility from canonical rules.
GreedyBaseline, BitmaskDP, CP-SAT Hanan/MultiPath	Reuse algorithms and interfaces after candidate completeness and geometry verification.
PortfolioHybrid, ALNS, repaired 8 operators	Reuse as reference/starting implementation; independently validate hub assignments and ring reuse.
LinUCB (8 arms, 24 features, Sherman–Morrison updates)	Reuse implementation and tests; reset learned statistics if state/action/reward semantics change.
PPO implementation, RTX 4050 CUDA, checkpoints	Archive as reproducible baseline; retrain only after corrected environment and positive headroom evidence.
FloorPlanChecker and 179 reported passing tests	Reuse regression suite, but independently implement oracle tests; pass counts alone do not establish specification correctness.
SQLite baseline cache and benchmark runner	Reuse mechanisms; invalidate entries with versioned geometry, checker, objective and solver hashes.
Historical 100-plan LinUCB result: J 23.77 vs 24.63 baseline	Retain as historical result, not definitive after a specification change; rerun paired benchmarks.

4. What went wrong or remains unproven

- Potentially incomplete room-door adjacency or candidate-zone generation may overestimate minimum hubs. The six-node/six-hub image is a trigger for investigation, not proof of a defect.
- “Exact optimal hub count” only holds relative to correctly enumerated feasible candidate zones and validated geometric placement/routing. It is not a global cable-cost proof.
- Prior reports used inconsistent interpretations (initial 20-node statement versus authoritative 5-node limit; possible confusion between cable traversal and zone connectivity).
- Original PPO learned a near no-op action (99.18% reported selection), entropy collapsed, and a five-plan validation set stayed flat. Reward design favored safe non-improvement.
- PPO environment stepping was sequential despite multiple configured environments; hidden validation steps and repeated routing caused ~12-hour pilot duration.
- Previously reported algorithm comparisons used differing sample sizes, feasible subsets and budgets; historical scores are not directly interchangeable.
- Infeasible/anomalous raw records were sometimes over-quarantined; 95–100 m^2 rooms are not inherently infeasible. Dataset partition accounting must be regenerated and reconciled.
- Visual overlays may not faithfully reflect underlying hub and ring JSON; visual correctness must be independently checked at the final stage.

5. Non-goals, principles and success criteria

- No UI, frontend, dashboard, styling or presentation deliverables until after final algorithm selection.
- No forced use of RL, GPU or a particular optimizer; choose based on unbiased cost, feasibility and runtime evidence.
- No weakening official checker or constraints to improve reported scores.

- No claim of global optimality without proof/certificate; distinguish proven H, best-found J, lower bounds and timeouts.

Success gates: (a) official-rule conformance on independent adversarial fixtures; (b) exact or explicitly bounded minimum H for benchmark-eligible plans; (c) valid closed rings and reproducible J; (d) strongest feasible J within a stated wall-clock budget; (e) robust held-out performance with uncertainty and failure accounting; (f) resumable evaluation across all raw records.

6. End-to-end phased implementation plan

Phase 0 — Freeze and provenance

Work: Snapshot repository, commit, old results and model states. Inventory raw inputs, JSON schemas, dataset IDs, version hashes, baseline cache and historical evaluation protocols.

Deliverables: Reproducibility manifest, asset register, old-vs-new rules diff.

Exit gate: Every artifact traceable; no destructive edits.

Phase 1 — Formal specification and independent oracle

Work: Translate slides 2–8 into machine-checkable invariants; test 5-node capacity, 100 m², fixed nodes, scale, contiguity, 0.5 m clearance, door-only traversal, ring closure and J. Build tiny handcrafted counterexamples.

Deliverables: Specification matrix, independent validator/oracle, property and negative tests.

Exit gate: Independent oracle and official checker agree on curated fixtures; disagreements documented.

Phase 2 — Dataset and geometry foundation

Work: Reparse all 17k; repair polygons deterministically; derive inner/free space, walls/windows, door gaps, eligible rooms and fixed node coordinates; separate traversability graph from zone adjacency; log ambiguous passages.

Deliverables: Versioned geometry IR, geometry tests, eligibility ledger and stratified split manifests.

Exit gate: 17k IDs reconciled exactly; units and door graphs pass geometric fixtures; no silent exclusions.

Phase 3 — Proven minimum-hub search

Work: Enumerate connected room subsets (up to documented small-N limits); filter area/node capacity; compute placement feasibility and route feasibility; solve minimum set partition with bitmask DP/CP-SAT; use counterexample-guided refinement for unrealizable candidates.

Deliverables: Certified H bounds, candidate-zone catalog, witnesses, infeasibility/timeout categories.

Exit gate: For small synthetic instances exhaustive reference agrees; suspicious six-node plan independently checked.

Phase 4 — Feasible closed-ring engine

Work: For each feasible zone, generate legal hub points, axis-aligned route graph through door gaps, node order (Held–Karp where practical), complete H-return ring; preserve alternative equal-H partitions.

Deliverables: Validated initial solutions, route certificates, deterministic fallback.

Exit gate: Independent oracle passes full JSON and polylines on representative layouts.

Phase 5 — Cost-optimal search portfolio

Work: Optimize J among equal-H feasible solutions using exact small-instance search, route rerouting, hub-location refinement, partition swaps, ALNS, CP-SAT where tractable and portfolio scheduling. Track best incumbent and lower bounds.

Deliverables: Portfolio of algorithms with shared interface and budget control.

Exit gate: Paired, equal-budget tests show best-found J and no H regression; timeouts reported.

Phase 6 — Learning research: LinUCB and PPO

Work: Audit corrected action space and improvement headroom; compare random, adaptive ALNS, LinUCB and PPO. Reuse old training examples only after semantic verification; regenerate invalid examples. Train shared policies on training partition, validate separately.

Deliverables: Learning baselines, ablations, trained/checkpointed models, reward and action diagnostics.

Exit gate: Learning method must outperform simpler baseline or offer proven speed/robustness benefit to be retained.

Phase 7 — Model/algorithm selection

Work: Freeze all configurations; repeated seeds, matched wall-clock and operator budgets, complexity-stratified validation; evaluate held-out test once for final claims; report paired J, H, feasibility, runtime and confidence intervals.

Deliverables: Champion selection report, statistical analysis and version-locked run manifest.

Exit gate: Chosen optimizer/portfolio wins by predefined lexicographic decision policy; no test-set tuning.

Phase 8 — Full 17,000-plan operational evaluation

Work: Run selected method over all raw IDs with deterministic status categories, per-plan output JSON, checker metrics, checkpoints, resume, resource telemetry and audit sampling. Keep held-out report separate.

Deliverables: Full results ledger, validated solution archive, failures, throughput and cost distribution.

Exit gate: Every input accounted for; each feasible output independently checked; no missing IDs.

Phase 9 — Final research consolidation

Work: Compare certified H, best-found J, ablation value of each solver, latency and generalization; publish limitations, reproducibility instructions and future work. UI is a separate subsequent project.

Deliverables: Technical report, reproducible code, final algorithm specification.

Exit gate: Independent rerun on representative subset reproduces key findings.

7. RL decision: train on all 17,000?

Not as one undifferentiated training corpus. The 17,000 records are the complete raw dataset, not 17,000 independent models. Create grouped, leakage-resistant training/validation/held-out partitions of eligible

plans. A shared LinUCB or PPO policy can encounter training plans repeatedly; validation informs selection; held-out test is never used for tuning. After selection, evaluate the frozen method operationally on all 17,000 raw records, including explicit invalid/infeasible statuses.

Recommended experiment ladder: (1) confirm nontrivial improvement headroom using all operators on a stratified sample; (2) random/adaptive ALNS baselines; (3) LinUCB online learning on training plans; (4) supervised operator pretraining only if labels are genuinely useful; (5) corrected PPO pilot with measured SPS, entropy, action diversity and paired J; (6) scale only if validation beats simpler methods. PPO training is optional evidence-based research, not a mandatory production dependency.

Reuse old PPO checkpoint only as historical comparison; do not resume it into changed observation/action/reward semantics. Reuse LinUCB update code, not necessarily its learned arm matrices. Old per-plan solutions can seed search only after revalidation with current geometry and objective; old labels/rewards must be recomputed if semantics changed.

8. Resource reuse policy: no compromises, no wasted work

Resource	Keep / regenerate rule
Raw ResPlan.pkl / official slide deck	Immutable canonical sources; always reuse.
Parsed geometry and adjacency	Reuse only if coordinate scaling, polygon repair, door interpretation and hash match current specification.
Baseline JSON and SQLite cache	Revalidate; key by plan, geometry, checker, objective, solver and parameters; invalidate stale entries.
ALNS operators, DP, CP-SAT, A*	Reuse tested implementation, but re-run correctness and comparative performance after foundational changes.
Historical benchmark CSVs	Archive for regression/history; never merge old scores into new claims without identical semantics and evaluation protocol.
LinUCB 24-feature encoder / bandit matrices	Reuse encoder if feature semantics stable; otherwise rederive and reset matrices. Relearn only on training split.
PPO policy weights and trajectories	Use as baseline or pretraining candidate only after schema and reward compatibility checks; otherwise train afresh.
GPU RTX 4050 and CPU i5-13450HX	CPU for geometry/ALNS/DP; GPU for batched PPO learning if justified. Profile end-to-end, not utilization alone.

9. Experimental design and decision rules

- Primary ranking is lexicographic: validity first; then H; then J; then area-gap tie-break. Never report “lower J wins” if H differs.
- Use same plans, initial incumbents, legal constraints, random seeds and matched wall-clock budgets; additionally report fixed-operator-budget experiments.
- Report feasibility denominators (raw, eligible, solvable, timeout, malformed) and per-plan paired differences, not only mean J.
- Report success rate for proving H, relative J improvement, median and P95 runtime, memory, operator no-op rate, and per-complexity results.
- Keep training/validation/test grouping immutable; online bandit updates in evaluation require separately labeled predeclared protocol. Frozen-policy evaluation is primary for generalization.
- For small synthetic instances use exhaustive independent solver as oracle; for large plans report lower bounds and optimality gaps where computable.

- Choose a portfolio if different solvers dominate different plan classes; dispatch rules must be learned on training/validation only and included in runtime accounting.

10. Parallel subagents and critical path

Run independent work concurrently but respect dependencies: Agent A formal spec/oracle; B geometry/data; C hub exact optimization; D route engine; E benchmarking/reproducibility; F learning research. A and B begin together; C/D use agreed interfaces and fixtures; E builds harness in parallel; F initially audits historical data and waits for validated environments before retraining. Use worktrees, file ownership, shared schema contracts, and a single heavy benchmark scheduler to avoid laptop contention.

The lead orchestrator owns integration and gate sign-off. Do not launch full training or all-17k runs automatically before Phase 2–5 correctness gates. Measure throughput on representative easy/medium/hard plans before estimating calendar time.

11. Risks, open questions and mitigations

- Door/passage representation may be ambiguous in ResPlan: produce geometric evidence and uncertain-case status; never silently invent connections.
- Official example indicates six nodes can use two hubs, but this does not establish the minimum for the earlier screenshot; identify that exact plan ID and produce a witness or infeasibility explanation.
- Slides state fixed node points; mentor says teams generate placements. Use detailed slide as canonical for benchmark inputs, document conflict and test input contract.
- Raw 17k contains oversized/anomalous plans; distinguish official filtered benchmark from full raw corpus and retain audit trail.
- “Optimal” J can be intractable: report proven hub optimality separately from heuristic routing-cost best-known.
- Potential data leakage from precomputed baselines or online adaptation: version cache and enforce partition controls.
- CPU-heavy route synthesis may dominate GPU training: use profiling, caching and bounded episodes before scaling.

12. Acceptance checklist and final deliverables

- Formal spec and independent validator with adversarial examples
- 17k raw-record reconciliation, eligibility ledger and frozen split manifests
- Geometry and door-connectivity graph verified on real and synthetic cases
- Minimum-hub solver with explicit proof status and witness solutions
- Complete legal closed-ring generator and official J recomputation
- Fair classical + ALNS + LinUCB + PPO research comparison
- Held-out evaluation with confidence intervals and no leakage
- Best algorithm/portfolio chosen by predefined criteria
- Resumable 17k operational run with every record accounted for
- Reproducible code, tests, configs, logs and final technical report
- UI/visual product layer deferred to a separate post-research phase